Harry Hariyanto

Data & AI Engineering Leader | Agentic AI for Decision Intelligence

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PROFESSIONAL PROFILE

Data & AI engineering leader with 8+ years at Southeast Asia's leading technology platforms (Shopee, Gojek, Vidio), building the full stack from data mart & platforms to ML in production and LLM agents. Grew and led Shopee Indonesia's BI & Data Engineering team from 4 to 14+ engineers, lifting self-service data mart adoption from ~5% to ~50% and governance coverage from 13% to 58%. Led 5 production ML systems and saved an estimated US$500K in H1 2025 with Automation, AI chatbots and MCP agent integrations. Now building decision intelligence: ML that decides, agents that act, and results you can measure. Deeply passionate about building robust, data-driven cultures and democratizing access to insights and AI enablement.

PROFESSIONAL EXPERIENCE

MailPost Australia **Sydney, 2025-2026**

AI Engineer (Part-time)

- Built n8n + LLM (Gemma, Ollama) workflows over MongoDB automating data processing and maintained social media data pipelines across Facebook, Instagram, X, and YouTube.

Shopee Indonesia **Jakarta, 2020-2025**

Assistant Manager – BI & Data Engineering

- **Leadership & Strategic Alignment:** Grew team from 4 to 14+ engineers. Tracked the progress of OKR quarterly and annual objectives. Bridged the gap between Business and Technology by establishing administrative Policies, Procedures, and Standards under the Business Intelligence Team. Collaborated with Local and Regional teams, supporting departments including BD, Marketing, Supply Chain, and Fintech. Discussed and laid out project specifications with HODs, Managers, Project Managers, and Engineering Leads. Selected for Shopee's Accelerated Development Program (2024).
- **Self-Service BI & Data Modelling:** Lay the foundation for data pipeline design across the Data Lake, Data Warehouse, and Data Mart. Designed dimensional models and standardised core business metrics, creating self-service datasets that increased data mart usage penetration across the organisation from ~5% to ~50%.
- **Data Governance & Security:** Spearheaded data governance projects to ensure all assets were documented. Increased data governance monitoring from 13% to 58% over H1 2025, implementing automated access approvals for stakeholders accessing sensitive data in alignment with ISO security standards.
- **Cross-Functional Analytics & Regulatory Compliance:** Engineered scalable data ecosystems to support diverse business arms, including Supply Chain, Marketing & Ads (loyalty, gamification, Shopee Ads, Shopee Live & KOL, P&L), and Business Development. Additionally supported the Fintech division (ShopeePay) by building highly secure, automated pipelines to deliver mandatory financial reports to government regulatory bodies (Bank Indonesia, OJK), demonstrating the ability to handle strict compliance within a high-velocity e-commerce environment.
- **Cost Reduction & Business Automation:** Designed, set up, and built robust automation systems using Airflow, Python, Django, Flask, and LLMs. Built AI chatbots and contributed to company-wide AI tools and MCP agent integration, saving an estimated ~$500K USD in manpower costs by automating manual business processes in H1 2025 alone. Optimized internal workflows with Python.
- **Infrastructure, Scale & Reliability:** Collaborating with the Data Infrastructure Team to architect data systems handling petabytes of data, reducing server costs by ~50% through algorithmic optimisation. Giving consultation for the architecture design, review, and migration from on-premise infrastructure to GCP cloud services.
- **AI & ML Deployment:** Led the engineering and collaborated with the Data Science Team, building end-to-end pipelines for OCR (Computer Vision), Fraud Detection, Logistics optimisation, User Profiling, Ticket Categorisation.
- **Real-Time Data Processing:** Utilised Apache Flink and ClickHouse (CK) for streaming data and delivering real-time analytics dashboards. Automated data extraction and social media metric scraping across platforms.

Gojek Indonesia **Jakarta, 2019 – 2020**

BI Engineering/Data Warehouse Engineer

- **Multi-Market Data Architecture:** Built unified data layers and a 'Single Source of Truth' to support a rapidly scaling international expansion squad. Seamlessly aligned complex data and business metrics across 4 international markets (Vietnam, Thailand, Singapore, and Indonesia), eliminating ad-hoc analysis reliance.
- **Data Pipeline & Governance Strategy:** Designed, maintained, and developed foundational frameworks for the central data warehouse, robust ETL pipelines, and data governance, ensuring reliable data availability for regional BI Analysts.
- **Enterprise System Integration:** Engineered and maintained data pipelines by successfully integrating crucial third-party enterprise platforms, including Salesforce, Asana, and Workday, into the central analytics ecosystem.

VIDIO (Vidio.com) **Jakarta, 2017 – 2019**

Data Engineer/Machine Learning Engineer

- Built and deployed a text classification model, reducing spam by 89%. Worked on Spam Detection and Recommendation Systems for Vidio.com and BBM using Flask and Pandas.
- Optimized recommendation algorithms to boost video play rates, successfully improving CTR by 2%.
- Created and managed robust ETLs on the GCP environment using Apache Spark, Scala, Cloud Dataproc, and Cloud Dataflow.

EDUCATION

Master of Data Science (ML & AI) *The University of Sydney | Sem 2 2025 – Dec 2026*

- **Research:** Research Student at the Sensors, Clouds, and Services Lab (SCS Lab) investigating Swarm-based Drone-as-a-Service (SDaaS) under the supervision of Prof Athman Bouguettaya.

Bachelor of Informatics (Computer Science) *Institut Teknologi Sepuluh Nopember (ITS) | 2013 – 2017*

- **GPA: 3.79/4.00 (Magna Cum Laude)**

RESEARCH & ACADEMIC PROJECTS

Publication: **Non-invasive Detection of Multilevel Diabetes (IEEE)** | *First Author, 2017*

- Proposed and engineered a novel signal processing pipeline using Wavelet Transforms for sensor data analysis | Garnered **27 citations** to date, validating long-term relevance in the field.

TECHNICAL & ANALYTICAL SKILLS

- **Analytical & BI:** SQL, Tableau, Looker, Dimensional Modelling, Metric Standardisation, KPI & Dashboard Design, Experimental Design
- **Leadership & Practice:** Team Leadership & Hiring, Data Governance, OKR Ownership, Architecture Design, Stakeholder Management, Agile Methodologies, Technical Writing
- **AI, ML & Agents:** Python (PyTorch, TensorFlow, Scikit-learn, XGBoost/LightGBM, Hugging Face, MLflow), Deep Learning, Reinforcement Learning, Computer Vision, LLMs, LangChain, LangGraph, LlamaIndex, MCP, Agent SDKs, Vector DBs, Langfuse/LangSmith, Multi-agent Systems
- **Data Engineering & Automation:** Apache Spark, Apache Flink, Kafka, Airflow, n8n, dbt, Cube (semantic layer)
- **Cloud & Infrastructure:** GCP, SeaCloud, Docker, Kubernetes, Ansible, CI/CD Pipelines, Terraform
- **Cloud Data Warehouses & DBMS:** BigQuery, ClickHouse, PostgreSQL, Hive, SQL, Cloud MemStore, MongoDB, DuckDB